A sharp bound for the least signless Laplacian eigenvalue of triangle-free graphs

Dongxiu Cai[1] Zhengbo Chen[1] Xiao-Dong Zhang[1,2]

Abstract

We prove that every finite simple triangle-free graph G on $n \geq 1$ vertices satisfies $q_{\min}(G) \leq 7n/50$, where $q_{\min}(G)$ is the least eigenvalue of its signless Laplacian matrix $Q_G = D_G + A_G$, with D_G the diagonal matrix of vertex degrees and A_G the adjacency matrix. The constant is sharp, since the Higman–Sims graph and its balanced blow-ups attain equality, as shown by de Lima, Nikiforov and Oliveira (2016). This proves the bound conjectured by Balogh, Clemen, Lidický, Norin and Volec (2023). We also prove that every graph attaining equality is isomorphic to a balanced blow-up of the Higman–Sims graph. The proof uses five test column vectors associated with an ordered edge to obtain a matrix Rayleigh inequality. Our use of local Rayleigh inequalities together with flag-algebra semidefinite programming is inspired by Zhou (2026). We use semidefinite programming and rational approximation to find the required positive semidefinite matrices, and complete the proof by exact correction and verification.

2020 Mathematics Subject Classification. Primary 05C50; Secondary 05C35, 90C22.

1 Introduction

Let G be a finite simple graph on $n \geq 1$ vertices, with adjacency matrix A_G. We write $q_{\min}(G)$ for the least eigenvalue of its signless Laplacian

$$Q_G = \mathrm{diag}_{v \in V(G)}(\deg_G(v)) + A_G,$$

where $\deg_G(v) = |\{w \in V(G) : \{v, w\} \in E(G)\}|$ is the degree of v, and $\mathrm{diag}_{z \in S}(b_z)$ denotes the diagonal matrix indexed by a finite set S, with entry b_z at (z, z) and zero at every off-diagonal position. For every real column vector u indexed by $V(G)$,

$$u^{\mathsf{T}} Q_G u = \sum_{\{v,w\} \in E(G)} (u_v + u_w)^2.$$

If G is bipartite, choosing u to be 1 on one part and -1 on the other gives a zero Rayleigh quotient, so $q_{\min}(G) = 0$ by the nonnegativity above. Conversely, if G is connected and $q_{\min}(G) = 0$, a nonzero zero-eigenvector u satisfies $u_v = -u_w$ on every edge; connectedness makes all coordinates nonzero, and their signs give a bipartition. Upper bounds on $q_{\min}(G)$ for triangle-free graphs are related to questions about sparse vertex sets and the number of edges that must be deleted to obtain a bipartite graph [1, 3].

[1] School of Mathematical Sciences, MOE-LSC, SHL-MAC, Shanghai Jiao Tong University, 800 Dongchuan Road, Shanghai 200240, P.R. China. Emails: diudiutse@sjtu.edu.cn (Dongxiu Cai) and czb911@sjtu.edu.cn (Zhengbo Chen).

[2] Corresponding author: Xiao-Dong Zhang. Email: xiaodong@sjtu.edu.cn.

Brandt [3] conjectured that every regular triangle-free graph of order n satisfies $q_{\min}(G) \le 4n/25$. For a d-regular graph, the degree matrix is dI, and therefore

$$q_{\min}(G) = d + \lambda_{\min}(A_G) = \lambda_{\max}(A_G) + \lambda_{\min}(A_G).$$

Consequently, Brandt's conjecture can also be formulated as a bound on the sum of the largest and smallest adjacency eigenvalues. Balogh, Clemen, Lidický, Norin and Volec [1] proved the stronger bound

$$q_{\min}(G) \le \frac{15}{94}n$$

for every triangle-free graph. Their argument uses a test column vector associated with an edge to bound the least eigenvalue by a ratio of induced-path counts, and then bounds that ratio by flag algebras.

The Higman–Sims graph has 100 vertices, is 22-regular and triangle-free, and has least adjacency eigenvalue -8. Its least signless Laplacian eigenvalue is therefore 14. In their concluding remarks, Balogh et al. [1] propose that $q_{\min}(G) \le 0.14n$ holds for every triangle-free graph. They also show that their edge test column vector gives the larger coefficient $42/275$ on the Higman–Sims graph. Improving only the subsequent path-count inequality cannot reach the conjectured coefficient using that same spectral reduction.

For a finite simple graph G and a positive integer m, denote by G^m its m-blow-up, obtained by replacing each vertex v by the independent set $\{(v,1),\dots,(v,m)\}$ and joining every copy of u to every copy of v if and only if $\{u,v\} \in E(G)$. Thus each edge of G is replaced by a complete bipartite graph between the corresponding sets of copies, and

$$\begin{aligned} V(G^m) &= V(G) \times \{1,\dots,m\}, \\ E(G^m) &= \{\{(u,i),(v,j)\} : \{u,v\} \in E(G),\ 1 \le i,j \le m\}. \end{aligned}$$

Denote by HS the Higman–Sims graph. In this paper, we prove the sharp bound conjectured by Balogh, Clemen, Lidický, Norin and Volec in 2023 [1], and characterize all graphs attaining equality, as stated in Theorem 1.

Theorem 1. *For every finite simple triangle-free graph G with $n = |V(G)| \ge 1$,*

$$q_{\min}(G) \le \frac{7}{50}n, \tag{1}$$

with equality if and only if $G \cong \mathrm{HS}^m$ for some positive integer m.

The spectral part of the proof uses five test column vectors for each ordered edge. Combining their Rayleigh inequalities produces two real symmetric 5×5 matrices: one records the averaged inner products of the test columns, and the other records their averaged signless Laplacian forms after subtracting the proposed bound. We construct a fixed positive definite matrix whose trace product with the second matrix is nonpositive. Its trace product with the first matrix is strictly positive when the graph has an edge. The matrix Rayleigh inequality then yields the desired eigenvalue bound. One of the test column vectors is obtained by applying a quadratic polynomial in the normalized adjacency matrix to the difference of two adjacency column vectors. This polynomial vanishes at the nonprincipal normalized adjacency eigenvalues of the equality example. The other column vectors also incorporate differences and local variation of vertex degrees.

To prove the trace inequality, we expand the matrix entries into rational linear combinations of graph homomorphism densities. Each term uses at most eight sampled positions. We then compare

coefficients on the complete list of 410 triangle-free graphs of order eight. The comparison expresses the required trace as the negative of a sum of nonnegative quadratic forms in conditional flag probabilities. These quadratic forms use explicit integer matrices whose positivity is verified by strict diagonal dominance. The remaining assertions are 410 exact finite identities. The accompanying data and programs specify the integer entries and check these identities using exact arithmetic.

The use of conditional flag probabilities follows the averaging principle of flag algebras [6]. Our use of local Rayleigh inequalities together with flag-algebra semidefinite programming is inspired by Zhou's work on $\{C_3, C_5\}$-free graphs. Zhou first averages local Rayleigh inequalities [8, Lemma 2.1, pp. 4–5], and then searches jointly for positive semidefinite matrices for the local Rayleigh expressions and ordinary flag squares, followed by rationalization and exact coefficient verification [8, Section 2.3, p. 7]. This is the method we draw on; the five test column vectors associated with an ordered edge and their 5×5 matrix constraint are constructed in the present paper. Here the eight positions are sampled independently with replacement from the original graph. Their sampled values may coincide, while the positions remain distinct vertices of the sampled graph. The resulting identities are exact for every $n \geq 1$, so an eight-vertex calculation proves the bound for all graph orders. We prove the required sampling identities directly.

Section 2 constructs the five test column vectors and the auxiliary matrices, and gives a sufficient condition for (1). Section 3 introduces the graph and flag densities and proves the sampling and positivity identities. Section 4 constructs B and proves (1) by exact verification of finitely many matrix identities. Section 5 characterizes the graphs attaining equality.

The programs and exact data used in the proof are available at
https://github.com/Diudiutse/triangle-free-signless-laplacian.

2 Construction of the auxiliary matrices and a sufficient condition for the upper bound

Throughout this section, let $G = (V, E(G))$ be a finite simple triangle-free graph with $n = |V| \geq 1$, and let $A = A_G$. Fix an ordering of V, and index the rows and columns of A by this ordering. For $v \in V$, its degree and the signless Laplacian are

$$\deg_G(v) = \big|\{w \in V : \{v, w\} \in E(G)\}\big| = \sum_{w \in V} A_{vw}, \qquad Q_G = \operatorname{diag}_{v \in V}(\deg_G(v)) + A.$$

For $C, D \subseteq V$, write

$$A_{C,D} = (A_{cd})_{c \in C,\, d \in D},$$

with rows and columns in the orders inherited from V. When $D = \{x\}$, write $A_{C,x}$ for $A_{C,\{x\}}$. In particular, $A_{V,x}$ is a real column vector indexed by V, with coordinate A_{zx} at $z \in V$. For a real column vector u indexed by V, write

$$\|u\|^2 = u^{\mathsf{T}} u = \sum_{z \in V} u(z)^2.$$

The notation I denotes the identity matrix of the order determined by context. An average over x, y always means independent uniform samples from V with replacement:

$$\mathbb{E}_{x,y} f(x, y) = \frac{1}{n^2} \sum_{x,y \in V} f(x, y).$$

Thus the ordered pairs are averaged with weight $1/n^2$, including pairs with $x = y$.

2.1 Definition of the auxiliary matrices

For every ordered pair $(x, y) \in V^2$, define the $n \times 5$ real matrix

$$U^{xy} = (U_0^{xy} \;\; U_1^{xy} \;\; U_2^{xy} \;\; U_3^{xy} \;\; U_4^{xy}), \tag{2}$$

whose columns are the following real column vectors indexed by V:

$$U_0^{xy} = \left(\frac{A}{n} - \frac{1}{50}I\right)(A_{V,x} - A_{V,y}), \tag{3}$$

$$U_1^{xy} = \frac{\deg_G(x) - \deg_G(y)}{n}(A_{V,x} + A_{V,y}), \tag{4}$$

$$U_2^{xy} = \mathrm{diag}_{z\in V}\left(\frac{\deg_G(z)}{n} - \frac{\deg_G(x) + \deg_G(y)}{2n}\right)(A_{V,x} - A_{V,y}), \tag{5}$$

$$U_3^{xy} = \left(\frac{A^2}{n^2} + \frac{3A}{50n} - \frac{1}{625}I\right)(A_{V,x} - A_{V,y}), \tag{6}$$

$$\begin{aligned} U_4^{xy} = {} & \mathrm{diag}_{z\in V}\left(\frac{\deg_G(z)}{n} - \frac{\deg_G(x) + \deg_G(y)}{2n}\right) U_0^{xy} \\ & + \frac{(\deg_G(x) - \deg_G(y))^2}{4n^2}(A_{V,x} - A_{V,y}). \end{aligned} \tag{7}$$

These definitions apply to all ordered pairs, including nonedges. The factor A_{xy} in the averages below makes nonedges contribute zero.

For $i, j \in \{0, 1, 2, 3, 4\}$, define

$$\mathsf{D}_{ij} = \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U_i^{xy})^{\mathsf{T}} U_j^{xy}\right], \tag{8}$$

$$\mathsf{F}_{ij} = \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U_i^{xy})^{\mathsf{T}}\left(\frac{Q_G}{n} - \frac{7}{50}I\right) U_j^{xy}\right], \tag{9}$$

and set

$$\mathsf{D} = (\mathsf{D}_{ij})_{i,j=0}^4, \qquad \mathsf{F} = (\mathsf{F}_{ij})_{i,j=0}^4. \tag{10}$$

Both matrices are 5×5, with rows and columns indexed by $0, 1, 2, 3, 4$ in the column order of U^{xy}. Equivalently,

$$\mathsf{D} = \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U^{xy})^{\mathsf{T}} U^{xy}\right], \tag{11}$$

$$\mathsf{F} = \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U^{xy})^{\mathsf{T}}\left(\frac{Q_G}{n} - \frac{7}{50}I\right) U^{xy}\right]. \tag{12}$$

The matrix Q_G is symmetric, so the two matrices in (10) are real symmetric matrices. The trace of a square matrix M is denoted by $\mathrm{tr}(M) = \sum_i M_{ii}$.

2.2 A sufficient condition for the upper bound

Lemma 1. *If there exists a real symmetric positive definite 5×5 matrix B such that, for every finite simple triangle-free graph G with $n = |V(G)| \geq 1$,*

$$\mathrm{tr}(B\mathsf{F}) \leq 0,$$

where F is defined from G by (12), then (1) holds for every finite simple triangle-free graph G of order $n \geq 1$.

Proof. Let B satisfy the hypotheses, and fix a finite simple triangle-free graph G with $n \geq 1$ vertices. If $E(G) = \varnothing$, then $Q_G = 0$ and $q_{\min}(G) = 0 \leq 7n/50$. We may therefore assume $E(G) \neq \varnothing$.

We first check that D is positive semidefinite and has positive trace. For any real column vector $u = (u_0, \ldots, u_4)^{\mathsf{T}} \in \mathbb{R}^5$, equations (2) and (8) give

$$\begin{aligned} u^{\mathsf{T}}\mathsf{D}u &= \sum_{i,j=0}^{4} u_i u_j \mathsf{D}_{ij} \\ &= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\sum_{i,j=0}^{4} u_i u_j (U_i^{xy})^{\mathsf{T}} U_j^{xy}\right] \\ &= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\|U^{xy}u\|^2\right] \geq 0. \end{aligned} \tag{13}$$

Here $A_{xy}/n \geq 0$ for every ordered pair, so D is positive semidefinite.

For an ordered edge (x, y), triangle-freeness implies $A_{xz}A_{yz} = 0$ for every $z \in V$. The coordinate at z in (3) is

$$U_0^{xy}(z) = \frac{1}{n}\sum_{t\in V} A_{zt}(A_{tx} - A_{ty}) - \frac{1}{50}(A_{zx} - A_{zy}). \tag{14}$$

In particular, using $A_{xx} = A_{yy} = 0$, $A_{xy} = A_{yx} = 1$, symmetry of A, and $A_{zt}^2 = A_{zt}$, we obtain

$$\begin{aligned} U_0^{xy}(x) &= \frac{1}{n}\sum_{t\in V}(A_{xt}^2 - A_{xt}A_{yt}) + \frac{1}{50} \\ &= \frac{\deg_G(x)}{n} + \frac{1}{50} > 0, \end{aligned} \tag{15}$$

$$\begin{aligned} U_0^{xy}(y) &= \frac{1}{n}\sum_{t\in V}(A_{yt}A_{xt} - A_{yt}^2) - \frac{1}{50} \\ &= -\frac{\deg_G(y)}{n} - \frac{1}{50} < 0. \end{aligned} \tag{16}$$

Thus $U_0^{xy} \neq 0$ for every ordered edge. Taking the trace in (11) yields

$$\begin{aligned} \operatorname{tr}(\mathsf{D}) &= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\sum_{i=0}^{4}\|U_i^{xy}\|^2\right] \\ &\geq \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\|U_0^{xy}\|^2\right] > 0. \end{aligned} \tag{17}$$

The strict inequality holds because the edge set is nonempty and every ordered edge has positive sampling probability $1/n^2$ and a nonzero column U_0^{xy}.

By the definition of the least eigenvalue, $Q_G - q_{\min}(G)I$ is positive semidefinite. Subtracting the corresponding multiple of (11) from (12), for every $u \in \mathbb{R}^5$ we obtain

$$\begin{aligned} &u^{\mathsf{T}}\left[\mathsf{F} - \left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)\mathsf{D}\right]u \\ &= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U^{xy}u)^{\mathsf{T}}\left(\frac{Q_G}{n} - \frac{7}{50}I - \left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)I\right)(U^{xy}u)\right] \\ &= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U^{xy}u)^{\mathsf{T}}\frac{Q_G - q_{\min}(G)I}{n}(U^{xy}u)\right] \geq 0. \end{aligned} \tag{18}$$

Hence $\mathsf{F} - (q_{\min}(G)/n - 7/50)\mathsf{D}$ is positive semidefinite. The trace of the product of two real symmetric positive semidefinite matrices is nonnegative [2, Example 2.24]. Applying this fact with the positive definite matrix B and the matrix in (18), and using linearity of the trace, gives

$$\begin{aligned} 0 &\le \operatorname{tr}\left(B\left[\mathsf{F} - \left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)\mathsf{D}\right]\right) \\ &= \operatorname{tr}(B\mathsf{F}) - \left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)\operatorname{tr}(B\mathsf{D}). \end{aligned} \tag{19}$$

To divide by $\operatorname{tr}(B\mathsf{D})$, we need its strict positivity. Since B is positive definite, $\lambda_{\min}(B) > 0$ and $B - \lambda_{\min}(B)I$ is positive semidefinite. Using (13) and the same trace nonnegativity gives

$$\operatorname{tr}(B\mathsf{D}) - \lambda_{\min}(B)\operatorname{tr}(\mathsf{D}) = \operatorname{tr}\left((B - \lambda_{\min}(B)I)\mathsf{D}\right) \ge 0.$$

Consequently, (17) implies

$$\operatorname{tr}(B\mathsf{D}) \ge \lambda_{\min}(B)\operatorname{tr}(\mathsf{D}) > 0. \tag{20}$$

Rearranging (19) and applying the hypothesis $\operatorname{tr}(B\mathsf{F}) \le 0$ now yields

$$\left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)\operatorname{tr}(B\mathsf{D}) \le \operatorname{tr}(B\mathsf{F}) \le 0.$$

Division by the positive quantity in (20) gives

$$\frac{q_{\min}(G)}{n} - \frac{7}{50} \le \frac{\operatorname{tr}(B\mathsf{F})}{\operatorname{tr}(B\mathsf{D})} \le 0,$$

and hence $q_{\min}(G) \le 7n/50$. Since G was arbitrary, this proves (1). □

In Sections 3 and 4, we construct a positive definite matrix B satisfying the condition in Lemma 1.

3 Definitions, notation, and preliminary lemmas

3.1 Graph maps, densities, and sampling

All graphs below are finite and simple. For an integer $s \ge 0$, write $[s] = \{1, \dots, s\}$, with $[0] = \varnothing$. The graph G denotes the ambient graph, with $n = |V(G)| \ge 1$.

For graphs J, H, define the set of graph isomorphisms by

$$\operatorname{Iso}(J,H) = \left\{\phi : V(J) \xrightarrow{\text{bij}} V(H) : (A_J)_{uv} = (A_H)_{\phi(u)\phi(v)} \text{ for all } u, v \in V(J)\right\}. \tag{21}$$

We write $J \cong H$ if $\operatorname{Iso}(J,H) \ne \varnothing$. For a subset $S \subseteq V(G)$, the induced subgraph is

$$G[S] = (S, \{\{u,v\} \in E(G) : u, v \in S\}).$$

The sets of homomorphisms, injective homomorphisms, and induced embeddings from J to G are, respectively,

$$\operatorname{Hom}(J,G) = \{\phi : V(J) \longrightarrow V(G) : \{\phi(u), \phi(v)\} \in E(G) \text{ for every } \{u,v\} \in E(J)\}, \tag{22}$$

$$\operatorname{Emb}(J,G) = \{\phi \in \operatorname{Hom}(J,G) : \phi \text{ is injective}\}, \tag{23}$$

$$\operatorname{Emb}_{\text{ind}}(J,G) = \left\{\phi : V(J) \hookrightarrow V(G) : (A_J)_{uv} = (A_G)_{\phi(u)\phi(v)} \text{ for all } u, v \in V(J)\right\}. \tag{24}$$

An injective homomorphism is required to preserve edges, whereas an induced embedding preserves both edges and nonedges.

Let $s = |V(J)|$. For $n \geq s$, write

$$(n)_s = n(n-1)\cdots(n-s+1) = \frac{n!}{(n-s)!}, \qquad (n)_0 = 1.$$

The homomorphism and injective homomorphism densities are

$$t(J,G) = \frac{|\operatorname{Hom}(J,G)|}{n^s}, \qquad t_{\mathrm{inj}}(J,G) = \frac{|\operatorname{Emb}(J,G)|}{(n)_s} \quad (n \geq s). \tag{25}$$

To write these densities as adjacency products, label $V(J)$ by $[s]$. Then

$$t(J,G) = \frac{1}{n^s} \sum_{x_1,\ldots,x_s \in V(G)} \prod_{\{a,b\} \in E(J)} (A_G)_{x_a x_b}, \tag{26}$$

$$t_{\mathrm{inj}}(J,G) = \frac{1}{(n)_s} \sum_{\substack{(x_1,\ldots,x_s) \in V(G)^s \\ x_a \neq x_b \text{ for } a \neq b}} \prod_{\{a,b\} \in E(J)} (A_G)_{x_a x_b}. \tag{27}$$

Each product is the indicator that the map $a \mapsto x_a$ preserves all edges of J; the second sum restricts these maps to injections. Empty products are taken to be 1. For $n \geq s$, the induced-subgraph density is

$$p(J,G) = \frac{\left|\{S \subseteq V(G) : |S| = s,\ G[S] \cong J\}\right|}{\binom{n}{s}}. \tag{28}$$

Sampling on positions. For an integer $h \geq 0$, let $\boldsymbol{X} = (X_1, \ldots, X_h)$ have independent uniform coordinates in $V(G)$. Thus $\boldsymbol{X}$ is uniformly distributed on $V(G)^h$, with replacement. Define the graph on the sampled positions by

$$G_{\boldsymbol{X}} = \left([h],\ \{\{a,b\} : 1 \leq a < b \leq h,\ \{X_a, X_b\} \in E(G)\}\right). \tag{29}$$

In particular,

$$(A_{G_{\boldsymbol{X}}})_{ab} = (A_G)_{X_a X_b} \qquad (a,b \in [h]). \tag{30}$$

Expectations and probabilities in this sampling rule are

$$\mathbb{E}_X f(\boldsymbol{X}) = \frac{1}{n^h} \sum_{\boldsymbol{X} \in V(G)^h} f(\boldsymbol{X}), \qquad \Pr(\boldsymbol{X} \in \mathcal{A}) = \frac{|\mathcal{A}|}{n^h} \quad (\mathcal{A} \subseteq V(G)^h). \tag{31}$$

For $s \in \{0, \ldots, 8\}$, let $\mathcal{G}_s$ be the set of all triangle-free graphs with vertex set $[s]$, that is,

$$\mathcal{G}_s = \left\{ ([s], E) : \begin{array}{l} E \subseteq \{\{a,b\} : 1 \leq a < b \leq s\}, \\ ([s], E) \text{ is triangle-free} \end{array} \right\}. \tag{32}$$

Clearly, graph isomorphism is an equivalence relation on $\mathcal{G}_s$. Let $\mathcal{T}_s \subseteq \mathcal{G}_s$ be a system of representatives for this equivalence relation; that is, for every $J \in \mathcal{G}_s$,

$$\left|\{H \in \mathcal{T}_s : J \cong H\}\right| = 1. \tag{33}$$

If G is triangle-free, then $G_{\boldsymbol{X}}$ is triangle-free. Consequently, for $0 \leq h \leq 8$,

$$V(G)^h = \bigsqcup_{H \in \mathcal{T}_h} \{\boldsymbol{X} \in V(G)^h : G_{\boldsymbol{X}} \cong H\}, \qquad \sum_{H \in \mathcal{T}_h} \Pr(G_{\boldsymbol{X}} \cong H) = 1. \tag{34}$$

The following identity relates homomorphism densities in the ambient graph to injective homomorphism densities in the sampled graphs.

Lemma 2. *Let G be a finite simple triangle-free graph of order $n \geq 1$, let $1 \leq h \leq 8$, and let J be a finite simple graph of order $s \leq h$, including $s = 0$. For a random tuple $\boldsymbol{X} = (X_1, \ldots, X_h) \in V(G)^h$, where the coordinates X_i are independent and uniformly distributed on $V(G)$, we have*

$$t(J, G) = \mathbb{E}_X t_{\mathrm{inj}}(J, G_{\boldsymbol{X}}) = \sum_{H \in \mathcal{T}_h} t_{\mathrm{inj}}(J, H) \Pr(G_{\boldsymbol{X}} \cong H). \tag{35}$$

Proof. Label $V(J)$ by $[s]$ and fix an injection $\phi : [s] \hookrightarrow [h]$. The random variables $X_{\phi(1)}, \ldots, X_{\phi(s)}$ are independent and uniform on $V(G)$, because ϕ selects different positions. Using (30) and (26), we obtain

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{X}} \prod_{\{a,b\} \in E(J)} (A_{G_{\boldsymbol{X}}})_{\phi(a)\phi(b)} &= \mathbb{E}_{\boldsymbol{X}} \prod_{\{a,b\} \in E(J)} (A_G)_{X_{\phi(a)} X_{\phi(b)}} \\
&= \frac{1}{n^s} \sum_{x_1, \ldots, x_s \in V(G)} \prod_{\{a,b\} \in E(J)} (A_G)_{x_a x_b} \\
&= t(J, G).
\end{aligned}$$

The graph $G_{\boldsymbol{X}}$ has h vertices, and there are $(h)_s$ injections from $[s]$ to $[h]$. By (27) and linearity of expectation,

$$\begin{aligned}
\mathbb{E}_X t_{\mathrm{inj}}(J, G_{\boldsymbol{X}}) &= \frac{1}{(h)_s} \sum_{\phi : [s] \hookrightarrow [h]} \mathbb{E}_{\boldsymbol{X}} \prod_{\{a,b\} \in E(J)} (A_{G_{\boldsymbol{X}}})_{\phi(a)\phi(b)} \\
&= \frac{1}{(h)_s} \sum_{\phi : [s] \hookrightarrow [h]} t(J, G) = t(J, G).
\end{aligned}$$

This proves the first equality in (35).

Since $t_{\mathrm{inj}}(J, \cdot)$ is invariant under graph isomorphism, the partition in (34) gives

$$\begin{aligned}
\mathbb{E}_X t_{\mathrm{inj}}(J, G_{\boldsymbol{X}}) &= \mathbb{E}_{\boldsymbol{X}} \left[\sum_{H \in \mathcal{T}_h} t_{\mathrm{inj}}(J, H) \mathbf{1}_{\{G_{\boldsymbol{X}} \cong H\}} \right] \\
&= \sum_{H \in \mathcal{T}_h} t_{\mathrm{inj}}(J, H) \mathbb{E}_{\boldsymbol{X}} \mathbf{1}_{\{G_{\boldsymbol{X}} \cong H\}} \\
&= \sum_{H \in \mathcal{T}_h} t_{\mathrm{inj}}(J, H) \Pr(G_{\boldsymbol{X}} \cong H).
\end{aligned}$$

This proves the second equality. □

3.2 Labeled graphs and the counting matrices

A type of order k is a triangle-free graph $\sigma = ([k], E(\sigma))$. A σ-labeled graph is a pair (J, θ), where J is a triangle-free graph and

$$\theta \in \mathrm{Emb}_{\mathrm{ind}}(\sigma, J).$$

The vertex $\theta(i)$ carries label i for each $i \in [k]$. For two σ-labeled graphs, define

$$(J, \theta) \cong_\sigma (J', \theta') \quad \Longleftrightarrow \quad \{\phi \in \mathrm{Iso}(J, J') : \phi \circ \theta = \theta'\} \neq \varnothing. \tag{36}$$

Thus an isomorphism of labeled graphs preserves each specified label. The relation $\cong_\sigma$ is an equivalence relation.

For $0 \le k \le \ell \le 8$, let $\iota : [k] \hookrightarrow [\ell]$ be the inclusion map $\iota(i) = i$. Define

$$\begin{aligned}\widehat{\mathcal{G}}_{\sigma,\ell} &= \{(J,\iota) : J \in \mathcal{G}_\ell,\ \iota \in \mathrm{Emb}_{\mathrm{ind}}(\sigma, J)\} \\ &= \{(J,\iota) : J \in \mathcal{G}_\ell,\ J[[k]] = \sigma\}. \end{aligned} \tag{37}$$

Here the first k vertices have labels $1, \dots, k$. Every σ-labeled graph with ℓ vertices is isomorphic to an element of (37): relabel $\theta(i)$ as i and give the remaining vertices the vertex names $k+1, \dots, \ell$. For the pairs in (37), the isomorphisms in (36) fix $1, \dots, k$ pointwise.

Let $\mathcal{F}_{\sigma,\ell} \subseteq \widehat{\mathcal{G}}_{\sigma,\ell}$ be a system of representatives for $\cong_\sigma$, and denote each representative pair by a single symbol P. Thus, for every $(J,\iota) \in \widehat{\mathcal{G}}_{\sigma,\ell}$,

$$\left|\{P \in \mathcal{F}_{\sigma,\ell} : (J,\iota) \cong_\sigma P\}\right| = 1. \tag{38}$$

We call the elements of $\mathcal{F}_{\sigma,\ell}$ the σ-flags of order ℓ. Each flag has k labeled vertices and $\ell - k$ unlabeled vertices.

For the counting matrices used below, take

$$\sigma \in \mathcal{T}_k, \qquad 0 \le k \le 6, \qquad \ell = \left\lfloor \frac{8+k}{2} \right\rfloor. \tag{39}$$

Then $2\ell - k \le 8$. Let H be a triangle-free graph with $|V(H)| = 8$, and let $P, P' \in \mathcal{F}_{\sigma,\ell}$. For a map $\rho : [8] \to V(H)$ and a set $S \subseteq [8]$, write $\rho(S) = \{\rho(i) : i \in S\}$ and write $\rho|_{[k]}$ for the restriction $i \mapsto \rho(i)$ on $[k]$. Define

$$C_H^\sigma(P,P') = \left| \left\{ \rho : \begin{array}{l} \rho : [8] \xrightarrow{\text{bij}} V(H), \\ (A_H)_{\rho(i)\rho(j)} = (A_\sigma)_{ij} \quad (1 \le i < j \le k), \\ (H[\rho([\ell])], \rho|_{[k]}) \cong_\sigma P, \\ (H[\rho([k] \cup \{\ell+1, \dots, 2\ell-k\})], \rho|_{[k]}) \cong_\sigma P' \end{array} \right\} \right|. \tag{40}$$

Figure 1 shows an example of a bijection ρ satisfying the conditions in (40). Fix an ordering of $\mathcal{F}_{\sigma,\ell}$ and use it for both the rows and the columns of the square matrix

$$C_H^\sigma = \left(C_H^\sigma(P,P')\right)_{P,P' \in \mathcal{F}_{\sigma,\ell}}.$$

Clearly, $C_H^\sigma(P,P') = C_H^\sigma(P',P)$ for all $P, P' \in \mathcal{F}_{\sigma,\ell}$, so C_H^σ is symmetric.

In (40), the two flags use the same labeled positions $[k]$ and the disjoint unlabeled position sets

$$\{k+1, \dots, \ell\}, \qquad \{\ell+1, \dots, 2\ell-k\}.$$

The remaining $8 - 2\ell + k$ positions are unused by either flag. The definition imposes no condition on edges between the two unlabeled groups or on edges incident with unused positions.

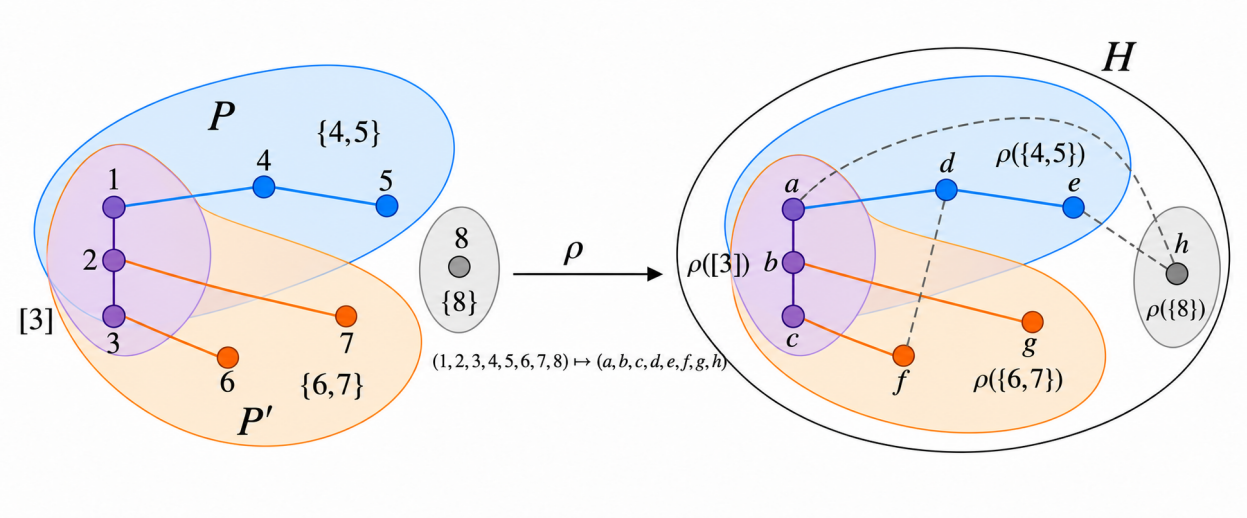


Figure 1: An admissible bijection for $k = 3$ and $\ell = 5$. The purple root path is σ; the blue and orange regions represent P and P', respectively, and position 8 is unused by either flag. The gray dashed edges ah, df, and eh are actual edges of H unconstrained by the two flags.

We now relate these counts to probabilities in the ambient graph G. Fix a triangle-free graph G of order $n \geq 1$. For any tuple $\boldsymbol{r} = (r_1, \ldots, r_k) \in V(G)^k$, let $G_{\boldsymbol{r}}$ be the graph on $[k]$ given by the same rule as (29); thus $(A_{G_{\boldsymbol{r}}})_{ij} = (A_G)_{r_i r_j}$. For $\boldsymbol{Y} = (Y_1, \ldots, Y_{\ell-k}) \in V(G)^{\ell-k}$, write

$$(\boldsymbol{r}, \boldsymbol{Y}) = (r_1, \ldots, r_k, Y_1, \ldots, Y_{\ell-k}).$$

For a flag $P \in \mathcal{F}_{\sigma,\ell}$, define

$$\Pr(P \mid \boldsymbol{r}) = \begin{cases} \dfrac{\left|\{\boldsymbol{Y} \in V(G)^{\ell-k} : (G_{(\boldsymbol{r},\boldsymbol{Y})}, \iota) \cong_\sigma P\}\right|}{n^{\ell-k}}, & G_{\boldsymbol{r}} = \sigma, \\ 0, & G_{\boldsymbol{r}} \neq \sigma. \end{cases} \tag{41}$$

When $G_{\boldsymbol{r}} = \sigma$, this is the probability that $(G_{(\boldsymbol{r},\boldsymbol{Y})}, \iota) \cong_\sigma P$, where the coordinates of $\boldsymbol{Y} = (Y_1, \ldots, Y_{\ell-k})$ are independent and uniformly distributed on $V(G)$. The map ι labels position i by i for $i \in [k]$. For a block of length zero, $V(G)^0$ consists of the unique empty tuple.

For $P, P' \in \mathcal{F}_{\sigma,\ell}$, define the event

$$\mathcal{A}^\sigma_{P,P'} = \left\{ \boldsymbol{x} = (x_1, \ldots, x_8) \in V(G)^8 : \begin{array}{l} G_{(x_1,\ldots,x_k)} = \sigma, \\ (G_{(x_1,\ldots,x_\ell)}, \iota) \cong_\sigma P, \\ (G_{(x_1,\ldots,x_k,x_{\ell+1},\ldots,x_{2\ell-k})}, \iota) \cong_\sigma P' \end{array} \right\}. \tag{42}$$

Lemma 3. *Let G be a finite simple triangle-free graph of order $n \geq 1$, and let $\boldsymbol{X} = (X_1, \ldots, X_8) \in V(G)^8$, where the coordinates X_i are independent and uniformly distributed on $V(G)$. Fix $\sigma \in \mathcal{T}_k$, with $0 \leq k \leq 6$, let $\ell = \lfloor (8+k)/2 \rfloor$, and set $\boldsymbol{r} = (X_1, \ldots, X_k)$. Then, for all $P, P' \in \mathcal{F}_{\sigma,\ell}$,*

$$\begin{aligned} \frac{1}{8!} \sum_{H \in \mathcal{T}_8} \Pr(G_{\boldsymbol{X}} \cong H) C^\sigma_H(P, P') &= \Pr\left(\boldsymbol{X} \in \mathcal{A}^\sigma_{P,P'}\right) \\ &= \mathbb{E}_{\boldsymbol{r}} \left[\mathbf{1}_{\{G_{\boldsymbol{r}} = \sigma\}} \Pr(P \mid \boldsymbol{r}) \Pr(P' \mid \boldsymbol{r}) \right]. \end{aligned} \tag{43}$$

Consequently,

$$M^\sigma_G := \sum_{H \in \mathcal{T}_8} \Pr(G_{\boldsymbol{X}} \cong H) C^\sigma_H \quad \textit{is positive semidefinite.} \tag{44}$$

Proof. Let $S_8 = \{\rho : [8] \xrightarrow{\text{bij}} [8]\}$. For $\boldsymbol{X} = (X_1, \dots, X_8) \in V(G)^8$ and $\rho \in S_8$, write

$$\boldsymbol{X} \circ \rho = (X_{\rho(1)}, \dots, X_{\rho(8)}).$$

Composition with a graph isomorphism gives a bijection between the orderings counted in (40), so $G_{\boldsymbol{X}} \cong H$ implies $C^\sigma_{G_{\boldsymbol{X}}} = C^\sigma_H$. By (40), (41), (42), and (34),

$$\begin{aligned}
&\frac{1}{8!}\sum_{H\in\mathcal{T}_8} \Pr(G_{\boldsymbol{X}} \cong H) C^\sigma_H(P,P') \\
&= \frac{1}{8!}\sum_{H\in\mathcal{T}_8} \mathbb{E}_{\boldsymbol{X}}\left[\mathbf{1}_{\{G_{\boldsymbol{X}}\cong H\}} C^\sigma_H(P,P')\right] \\
&= \frac{1}{8!}\mathbb{E}_{\boldsymbol{X}}\left[\sum_{H\in\mathcal{T}_8} \mathbf{1}_{\{G_{\boldsymbol{X}}\cong H\}} C^\sigma_H(P,P')\right] \\
&= \frac{1}{8!}\mathbb{E}_{\boldsymbol{X}} C^\sigma_{G_{\boldsymbol{X}}}(P,P') \\
&= \frac{1}{8!\, n^8}\sum_{\boldsymbol{X}\in V(G)^8} \left|\left\{\rho \in S_8 : \boldsymbol{X}\circ\rho \in \mathcal{A}^\sigma_{P,P'}\right\}\right| \\
&= \frac{1}{8!\, n^8}\left|\left\{(\boldsymbol{X},\rho) \in V(G)^8 \times S_8 : \boldsymbol{X}\circ\rho \in \mathcal{A}^\sigma_{P,P'}\right\}\right| \\
&= \frac{1}{8!\, n^8}\sum_{\boldsymbol{Y}\in\mathcal{A}^\sigma_{P,P'}} \left|\left\{(\boldsymbol{X},\rho) \in V(G)^8 \times S_8 : \boldsymbol{X}\circ\rho = \boldsymbol{Y}\right\}\right| \\
&= \frac{1}{8!\, n^8}\sum_{\boldsymbol{Y}\in\mathcal{A}^\sigma_{P,P'}} \sum_{\rho\in S_8} \underbrace{\left|\left\{\boldsymbol{X} \in V(G)^8 : \boldsymbol{X} = \boldsymbol{Y}\circ\rho^{-1}\right\}\right|}_{=1} \\
&= \frac{1}{8!\, n^8}\sum_{\boldsymbol{Y}\in\mathcal{A}^\sigma_{P,P'}} \sum_{\rho\in S_8} 1 \\
&= \frac{1}{8!\, n^8}\sum_{\boldsymbol{Y}\in\mathcal{A}^\sigma_{P,P'}} 8! \\
&= \frac{|\mathcal{A}^\sigma_{P,P'}|}{n^8} \\
&= \frac{1}{n^8}\left|\left\{\boldsymbol{Y} = (Y_1,\dots,Y_8) \in V(G)^8 : \begin{array}{l} G_{(Y_1,\dots,Y_k)} = \sigma, \\ (G_{(Y_1,\dots,Y_\ell)},\iota) \cong_\sigma P, \\ (G_{(Y_1,\dots,Y_k,Y_{\ell+1},\dots,Y_{2\ell-k})},\iota) \cong_\sigma P' \end{array}\right\}\right| \\
&= \frac{1}{n^8}\sum_{Y_1,\dots,Y_k\in V(G)}\ \sum_{Y_{k+1},\dots,Y_\ell\in V(G)}\ \sum_{Y_{\ell+1},\dots,Y_{2\ell-k}\in V(G)}\ \sum_{Y_{2\ell-k+1},\dots,Y_8\in V(G)} \\
&\qquad \cdot \mathbf{1}\left\{\begin{array}{l} G_{(Y_1,\dots,Y_k)} = \sigma, \\ (G_{(Y_1,\dots,Y_\ell)},\iota) \cong_\sigma P, \\ (G_{(Y_1,\dots,Y_k,Y_{\ell+1},\dots,Y_{2\ell-k})},\iota) \cong_\sigma P' \end{array}\right\} \\
&= \frac{1}{n^8}\sum_{Y_1,\dots,Y_k\in V(G)}\ \sum_{Y_{k+1},\dots,Y_\ell\in V(G)}\ \sum_{Y_{\ell+1},\dots,Y_{2\ell-k}\in V(G)}\ \sum_{Y_{2\ell-k+1},\dots,Y_8\in V(G)} \\
&\qquad \cdot \mathbf{1}_{\{G_{(Y_1,\dots,Y_k)}=\sigma\}} \mathbf{1}_{\{(G_{(Y_1,\dots,Y_\ell)},\iota)\cong_\sigma P\}} \\
&\qquad \cdot \mathbf{1}_{\{(G_{(Y_1,\dots,Y_k,Y_{\ell+1},\dots,Y_{2\ell-k})},\iota)\cong_\sigma P'\}}
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{n^8} \sum_{Y_1,\dots,Y_k \in V(G)} \mathbf{1}_{\{G_{(Y_1,\dots,Y_k)}=\sigma\}} \\
&\quad \cdot \sum_{Y_{k+1},\dots,Y_\ell \in V(G)} \sum_{Y_{\ell+1},\dots,Y_{2\ell-k} \in V(G)} \sum_{Y_{2\ell-k+1},\dots,Y_8 \in V(G)} \\
&\quad \cdot \mathbf{1}_{\{(G_{(Y_1,\dots,Y_\ell)},\iota) \cong_\sigma P\}} \mathbf{1}_{\{(G_{(Y_1,\dots,Y_k,Y_{\ell+1},\dots,Y_{2\ell-k})},\iota) \cong_\sigma P'\}} \\
&= \frac{1}{n^8} \sum_{\substack{Y_1,\dots,Y_k \in V(G) \\ G_{(Y_1,\dots,Y_k)}=\sigma}} \sum_{Y_{k+1},\dots,Y_\ell \in V(G)} \sum_{Y_{\ell+1},\dots,Y_{2\ell-k} \in V(G)} \sum_{Y_{2\ell-k+1},\dots,Y_8 \in V(G)} \\
&\quad \cdot \mathbf{1}_{\{(G_{(Y_1,\dots,Y_\ell)},\iota) \cong_\sigma P\}} \mathbf{1}_{\{(G_{(Y_1,\dots,Y_k,Y_{\ell+1},\dots,Y_{2\ell-k})},\iota) \cong_\sigma P'\}} \\
&= \frac{1}{n^8} \sum_{\substack{Y_1,\dots,Y_k \in V(G) \\ G_{(Y_1,\dots,Y_k)}=\sigma}} \left(\sum_{Y_{k+1},\dots,Y_\ell \in V(G)} \mathbf{1}_{\{(G_{(Y_1,\dots,Y_\ell)},\iota) \cong_\sigma P\}} \right) \\
&\quad \cdot \left(\sum_{Y_{\ell+1},\dots,Y_{2\ell-k} \in V(G)} \mathbf{1}_{\{(G_{(Y_1,\dots,Y_k,Y_{\ell+1},\dots,Y_{2\ell-k})},\iota) \cong_\sigma P'\}} \right) \left(\sum_{Y_{2\ell-k+1},\dots,Y_8 \in V(G)} 1 \right) \\
&= \frac{1}{n^8} \sum_{\substack{Y_1,\dots,Y_k \in V(G) \\ G_{(Y_1,\dots,Y_k)}=\sigma}} (n^{\ell-k} \Pr(P \mid (Y_1,\dots,Y_k))) \\
&\quad \cdot (n^{\ell-k} \Pr(P' \mid (Y_1,\dots,Y_k))) n^{8-2\ell+k} \\
&= \frac{1}{n^k} \sum_{Y_1,\dots,Y_k \in V(G)} \mathbf{1}_{\{G_{(Y_1,\dots,Y_k)}=\sigma\}} \Pr(P \mid (Y_1,\dots,Y_k)) \Pr(P' \mid (Y_1,\dots,Y_k)) \\
&= \mathbb{E}_{\boldsymbol{r}} \left[\mathbf{1}_{\{G_{\boldsymbol{r}}=\sigma\}} \Pr(P \mid \boldsymbol{r}) \Pr(P' \mid \boldsymbol{r}) \right].
\end{aligned}
$$

This proves (43). For any $\boldsymbol{w} = (w_P)_{P \in \mathcal{F}_{\sigma,\ell}} \in \mathbb{R}^{\mathcal{F}_{\sigma,\ell}}$, we have

$$
\begin{aligned}
&\boldsymbol{w}^{\mathsf{T}} M_G^\sigma \boldsymbol{w} \\
&= \sum_{P,P' \in \mathcal{F}_{\sigma,\ell}} w_P w_{P'} \left(\sum_{H \in \mathcal{T}_8} \Pr(G_{\boldsymbol{X}} \cong H) C_H^\sigma(P,P') \right) \\
&= 8! \sum_{P,P' \in \mathcal{F}_{\sigma,\ell}} w_P w_{P'} \mathbb{E}_{\boldsymbol{r}} \left[\mathbf{1}_{\{G_{\boldsymbol{r}}=\sigma\}} \Pr(P \mid \boldsymbol{r}) \Pr(P' \mid \boldsymbol{r}) \right] \\
&= 8! \, \mathbb{E}_{\boldsymbol{r}} \left[\mathbf{1}_{\{G_{\boldsymbol{r}}=\sigma\}} \sum_{P,P' \in \mathcal{F}_{\sigma,\ell}} w_P w_{P'} \Pr(P \mid \boldsymbol{r}) \Pr(P' \mid \boldsymbol{r}) \right] \\
&= 8! \, \mathbb{E}_{\boldsymbol{r}} \left[\mathbf{1}_{\{G_{\boldsymbol{r}}=\sigma\}} \left(\sum_{P \in \mathcal{F}_{\sigma,\ell}} w_P \Pr(P \mid \boldsymbol{r}) \right)^2 \right] \geq 0.
\end{aligned}
$$

Hence M_G^σ is positive semidefinite. □

4 Proof of the upper bound

By Lemma 1, it suffices to construct a fixed real symmetric positive definite 5×5 matrix B such that $\operatorname{tr}(B\mathsf{F}) \leq 0$ for every finite simple triangle-free graph G of order $n \geq 1$. We first express F as

an average of fixed matrices indexed by $\mathcal{T}_8$, and then use Lemma 3 to give finite matrix identities that imply this trace inequality.

4.1 Representation of F

Throughout this subsection, let G be a finite simple triangle-free graph, let $V = V(G)$ and $n = |V| \geq 1$, and write $A = A_G$. All positions listed in an expectation subscript are sampled independently and uniformly from V. For each $z \in V$,

$$\deg_G(z) = \sum_{t \in V} A_{zt} = n\mathbb{E}_t A_{zt}. \tag{45}$$

For fixed $x, y, z \in V$, substituting this identity and the entries of A^2 into (3)–(7) gives

$$\begin{aligned} U_0^{xy}(z) &= \mathbb{E}_s[A_{zs}(A_{sx} - A_{sy})] - \frac{1}{50}(A_{zx} - A_{zy}) \\ &= \mathbb{E}_s[A_{zs}A_{sx}] - \mathbb{E}_s[A_{zs}A_{sy}] - \frac{1}{50}A_{zx} + \frac{1}{50}A_{zy}. \end{aligned} \tag{46}$$

$$\begin{aligned} U_1^{xy}(z) &= \mathbb{E}_s[(A_{xs} - A_{ys})(A_{zx} + A_{zy})] \\ &= \mathbb{E}_s[A_{xs}A_{zx}] + \mathbb{E}_s[A_{xs}A_{zy}] - \mathbb{E}_s[A_{ys}A_{zx}] - \mathbb{E}_s[A_{ys}A_{zy}]. \end{aligned} \tag{47}$$

$$\begin{aligned} U_2^{xy}(z) &= \mathbb{E}_s\left[\left(A_{zs} - \frac{A_{xs} + A_{ys}}{2}\right)(A_{zx} - A_{zy})\right] \\ &= \mathbb{E}_s[A_{zs}A_{zx}] - \mathbb{E}_s[A_{zs}A_{zy}] - \frac{1}{2}\mathbb{E}_s[A_{xs}A_{zx}] + \frac{1}{2}\mathbb{E}_s[A_{xs}A_{zy}] \\ &\quad - \frac{1}{2}\mathbb{E}_s[A_{ys}A_{zx}] + \frac{1}{2}\mathbb{E}_s[A_{ys}A_{zy}]. \end{aligned} \tag{48}$$

$$\begin{aligned} U_3^{xy}(z) &= \mathbb{E}_{s,t}[A_{zs}A_{st}(A_{tx} - A_{ty})] \\ &\quad + \frac{3}{50}\mathbb{E}_s[A_{zs}(A_{sx} - A_{sy})] - \frac{1}{625}(A_{zx} - A_{zy}) \\ &= \mathbb{E}_{s,t}[A_{zs}A_{st}A_{tx}] - \mathbb{E}_{s,t}[A_{zs}A_{st}A_{ty}] \\ &\quad + \frac{3}{50}\mathbb{E}_s[A_{zs}A_{sx}] - \frac{3}{50}\mathbb{E}_s[A_{zs}A_{sy}] - \frac{1}{625}A_{zx} + \frac{1}{625}A_{zy}. \end{aligned} \tag{49}$$

$$\begin{aligned} U_4^{xy}(z) &= \mathbb{E}_{s,t}\left[\left(A_{zs} - \frac{A_{xs} + A_{ys}}{2}\right)\left(A_{zt}(A_{tx} - A_{ty}) - \frac{1}{50}(A_{zx} - A_{zy})\right)\right. \\ &\qquad \left. + \frac{1}{4}(A_{xs} - A_{ys})(A_{xt} - A_{yt})(A_{zx} - A_{zy})\right] \\ &= \mathbb{E}_{s,t}[A_{zs}A_{zt}A_{tx}] - \mathbb{E}_{s,t}[A_{zs}A_{zt}A_{ty}] - \frac{1}{50}\mathbb{E}_s[A_{zs}A_{zx}] + \frac{1}{50}\mathbb{E}_s[A_{zs}A_{zy}] \\ &\quad - \frac{1}{2}\mathbb{E}_{s,t}[A_{xs}A_{zt}A_{tx}] + \frac{1}{2}\mathbb{E}_{s,t}[A_{xs}A_{zt}A_{ty}] - \frac{1}{2}\mathbb{E}_{s,t}[A_{ys}A_{zt}A_{tx}] + \frac{1}{2}\mathbb{E}_{s,t}[A_{ys}A_{zt}A_{ty}] \\ &\quad + \frac{1}{100}\mathbb{E}_s[A_{xs}A_{zx}] - \frac{1}{100}\mathbb{E}_s[A_{xs}A_{zy}] + \frac{1}{100}\mathbb{E}_s[A_{ys}A_{zx}] - \frac{1}{100}\mathbb{E}_s[A_{ys}A_{zy}] \\ &\quad + \frac{1}{4}\mathbb{E}_{s,t}[A_{xs}A_{xt}A_{zx}] - \frac{1}{4}\mathbb{E}_{s,t}[A_{xs}A_{xt}A_{zy}] - \frac{1}{4}\mathbb{E}_{s,t}[A_{xs}A_{yt}A_{zx}] + \frac{1}{4}\mathbb{E}_{s,t}[A_{xs}A_{yt}A_{zy}] \\ &\quad - \frac{1}{4}\mathbb{E}_{s,t}[A_{ys}A_{xt}A_{zx}] + \frac{1}{4}\mathbb{E}_{s,t}[A_{ys}A_{xt}A_{zy}] + \frac{1}{4}\mathbb{E}_{s,t}[A_{ys}A_{yt}A_{zx}] - \frac{1}{4}\mathbb{E}_{s,t}[A_{ys}A_{yt}A_{zy}]. \end{aligned} \tag{50}$$

Expanding (9) in coordinates and using (45), we obtain, for $0 \le i, j \le 4$,

$$\begin{aligned}\mathsf{F}_{ij} &= \mathbb{E}_{x,y,z}\left[A_{xy}\frac{\deg_G(z)}{n}U_i^{xy}(z)U_j^{xy}(z)\right] \\ &\quad + \mathbb{E}_{x,y,z,t}\left[A_{xy}A_{zt}U_i^{xy}(z)U_j^{xy}(t)\right] - \frac{7}{50}\mathbb{E}_{x,y,z}\left[A_{xy}U_i^{xy}(z)U_j^{xy}(z)\right] \\ &= \mathbb{E}_{x,y,z,t}\left[A_{xy}A_{zt}U_i^{xy}(z)U_j^{xy}(z)\right] \\ &\quad + \mathbb{E}_{x,y,z,t}\left[A_{xy}A_{zt}U_i^{xy}(z)U_j^{xy}(t)\right] - \frac{7}{50}\mathbb{E}_{x,y,z}\left[A_{xy}U_i^{xy}(z)U_j^{xy}(z)\right]. \end{aligned} \tag{51}$$

By linearity of expectation and the distributive law, F_{ij} is a rational linear combination of terms of the form

$$\mathbb{E}_{v_1,\dots,v_s}\left[\prod_{r=1}^{m} A_{v_{a_r}v_{b_r}}\right], \qquad 0 \le s \le 8, \quad 0 \le m \le 8, \quad a_r, b_r \in [s]. \tag{52}$$

Here $v_1, \dots, v_s$ name distinct sampled positions. Their values in V need not be distinct.

The adjacency entries satisfy

$$A_{uv} = A_{vu}, \qquad A_{uv}^2 = A_{uv}, \qquad A_{uu} = 0, \qquad A_{uv}A_{vw}A_{wu} = 0.$$

The last equality uses triangle-freeness. Hence repeated adjacency factors can be replaced by one factor, and a term containing a loop or a triangle vanishes. For each remaining term in (52), set

$$J = \left([s], \{\{a_r, b_r\} : 1 \le r \le m\}\right).$$

Then J is triangle-free, and

$$\begin{aligned}\mathbb{E}_{v_1,\dots,v_s}\left[\prod_{r=1}^{m} A_{v_{a_r}v_{b_r}}\right] &= \frac{1}{n^s}\sum_{v_1,\dots,v_s \in V}\prod_{\{a,b\}\in E(J)} A_{v_a v_b} \\ &= t(J, G).\end{aligned}$$

Collecting the coefficients of isomorphic graphs therefore gives

$$\mathsf{F}_{ij} = \sum_{s=0}^{8}\sum_{J\in\mathcal{T}_s} c_{i,j,J}\, t(J, G), \qquad c_{i,j,J} \in \mathbb{Q}. \tag{53}$$

$c_{i,j,J}$ is independent of G. Interchanging i, j in the first and last terms of (51), and also interchanging z, t in its second term, gives $c_{i,j,J} = c_{j,i,J}$ after relabeling the sampled positions.

4.2 Construction of B

For each $H \in \mathcal{T}_8$, define the rational symmetric 5×5 matrix F_H by

$$(F_H)_{ij} = \sum_{s=0}^{8}\sum_{J\in\mathcal{T}_s} c_{i,j,J}\, t_{\mathrm{inj}}(J, H), \qquad 0 \le i, j \le 4. \tag{54}$$

Lemma 4. *For $k = 0, \dots, 6$, let $\ell = \lfloor (8+k)/2 \rfloor$. Suppose there exists a fixed real symmetric positive definite 5×5 matrix B, and for every type $\sigma \in \bigcup_{k=0}^{6}\mathcal{T}_k$ there exists a fixed real symmetric positive semidefinite matrix M_σ, with rows and columns indexed by $\mathcal{F}_{\sigma,\ell}$, such that*

$$\operatorname{tr}(BF_H) = -\sum_{k=0}^{6}\sum_{\sigma\in\mathcal{T}_k}\operatorname{tr}(M_\sigma C_H^\sigma) \qquad \text{for every } H \in \mathcal{T}_8. \tag{55}$$

Then $\operatorname{tr}(B\mathsf{F}) \le 0$ for every finite simple triangle-free graph G of order $n \ge 1$, where F is defined from G by (12). Consequently, (1) holds for every finite simple triangle-free graph G of order $n \ge 1$.

Proof. Fix a finite simple triangle-free graph G of order $n \geq 1$, and write $V = V(G)$. Let $\boldsymbol{X} = (X_1, \ldots, X_8) \in V^8$, where the coordinates X_i are independent and uniformly distributed on V. By Lemma 2,

$$\begin{aligned}
\mathsf{F}_{ij} &= \sum_{s=0}^{8} \sum_{J \in \mathcal{T}_s} c_{i,j,J} \sum_{H \in \mathcal{T}_8} t_{\mathrm{inj}}(J, H) \Pr(G_{\boldsymbol{X}} \cong H) \\
&= \sum_{H \in \mathcal{T}_8} \left(\sum_{s=0}^{8} \sum_{J \in \mathcal{T}_s} c_{i,j,J}\, t_{\mathrm{inj}}(J, H) \right) \Pr(G_{\boldsymbol{X}} \cong H) \\
&= \sum_{H \in \mathcal{T}_8} (F_H)_{ij} \Pr(G_{\boldsymbol{X}} \cong H).
\end{aligned}$$

Thus

$$\mathsf{F} = \sum_{H \in \mathcal{T}_8} \Pr(G_{\boldsymbol{X}} \cong H) F_H. \tag{56}$$

By (56), (55), linearity of the trace, and the positive semidefiniteness of M_G^σ for each type σ in (44), we obtain

$$\begin{aligned}
\operatorname{tr}(B\mathsf{F}) &= \sum_{H \in \mathcal{T}_8} \Pr(G_{\boldsymbol{X}} \cong H) \operatorname{tr}(BF_H) \\
&= -\sum_{H \in \mathcal{T}_8} \Pr(G_{\boldsymbol{X}} \cong H) \sum_{k=0}^{6} \sum_{\sigma \in \mathcal{T}_k} \operatorname{tr}(M_\sigma C_H^\sigma) \\
&= -\sum_{k=0}^{6} \sum_{\sigma \in \mathcal{T}_k} \operatorname{tr} \left(M_\sigma \sum_{H \in \mathcal{T}_8} \Pr(G_{\boldsymbol{X}} \cong H) C_H^\sigma \right) \\
&= -\sum_{k=0}^{6} \sum_{\sigma \in \mathcal{T}_k} \operatorname{tr}(M_\sigma M_G^\sigma) \leq 0.
\end{aligned} \tag{57}$$

Since G was arbitrary, Lemma 1 now proves (1). □

To find B and M_σ satisfying (55), we use semidefinite programming as follows:

$$\begin{aligned}
&\text{minimize} && \tau \\
&\text{subject to} && B_{00} = 1, \qquad B - 10^{-8} I \succeq 0, \qquad \tau \geq 0, \\
& && M_\sigma \succeq 0 \quad \left(\sigma \in \bigcup_{k=0}^{6} \mathcal{T}_k \right), \\
& && -\tau \leq 10^6 \left(\operatorname{tr}(BF_H) + \sum_{k=0}^{6} \sum_{\sigma \in \mathcal{T}_k} \operatorname{tr}(M_\sigma C_H^\sigma) \right) \leq \tau \quad (H \in \mathcal{T}_8).
\end{aligned} \tag{58}$$

Here $\succeq 0$ means positive semidefinite. We then apply rational approximation and exact correction to obtain rational positive semidefinite matrices B and M_σ satisfying

$$\operatorname{tr}(BF_H) + \sum_{k=0}^{6} \sum_{\sigma \in \mathcal{T}_k} \operatorname{tr}(M_\sigma C_H^\sigma) = 0 \qquad (H \in \mathcal{T}_8).$$

We store the resulting matrices through integer data $L, q, X_*, X_\sigma, V_\sigma, \Delta, a_\sigma$, where X_* and X_σ are symmetric and q, Δ, a_σ are positive. These data can be found in `programs/spectral_data.py`

and programs/sigma_data.py. The matrices required in Lemma 4 are

$$B = L(qI + X_*)L^{\mathsf{T}}, \qquad M_\sigma = \frac{2^{80}}{8!\,a_\sigma^2} V_\sigma (qI + X_\sigma) V_\sigma^{\mathsf{T}}, \tag{59}$$

The exact data satisfy the following conditions:

(i) L is invertible.

(ii) For every $i \in \{0, \dots, 4\}$,

$$q + (X_*)_{ii} > \sum_{j \neq i} |(X_*)_{ij}|.$$

(iii) For every $\sigma \in \bigcup_{k=0}^{6} \mathcal{T}_k$ and every row i of X_σ,

$$q + (X_\sigma)_{ii} > \sum_{j \neq i} |(X_\sigma)_{ij}|.$$

(iv) For every $H \in \mathcal{T}_8$,

$$\begin{aligned} 0 = {} & \frac{\Delta}{2^{80}} \operatorname{tr}\left(L(qI + X_*)L^{\mathsf{T}} F_H\right) \\ & + \sum_{k=0}^{6} \sum_{\sigma \in \mathcal{T}_k} \frac{\Delta}{8!\,a_\sigma^2} \operatorname{tr}\left(V_\sigma (qI + X_\sigma) V_\sigma^{\mathsf{T}} C_H^\sigma\right). \end{aligned} \tag{60}$$

Condition (i) follows directly since L is lower triangular with nonzero diagonal entries. Conditions (ii) and (iii) require 1 650 row inequalities, and condition (iv) requires 410 identities. The whole verification involves on the order of 10^{10} scalar arithmetic operations. The verification programs and instructions are given in programs/ and its README.md.

Proof of the inequality in Theorem 1. By conditions (i), (ii), and (iii) and the Gershgorin circle theorem [7], B is positive definite and every M_σ is positive semidefinite. Condition (iv), multiplied by $2^{80}/\Delta$, gives (55). Hence Lemma 4 proves (1). □

5 Equality cases

We now determine all graphs attaining equality in (1). The argument uses two additional exact identities for the matrices V_σ, followed by a count of common neighbors.

Lemma 5. *Let $k \in \{1, 2\}$, $\ell = \lfloor (8+k)/2 \rfloor$, and $\sigma = ([k], \varnothing)$. Every flag $P \in \mathcal{F}_{\sigma,\ell}$ has three unlabeled vertices. Define*

$$d(P) = \big|\{v \in V(P) \setminus [k] : \{1, v\} \in E(P)\}\big|.$$

For $k = 2$, also define

$$c(P) = \big|\{v \in V(P) \setminus [2] : \{1, v\} \in E(P),\ \{2, v\} \in E(P)\}\big|.$$

For the fixed matrices V_σ used in (59), the following statements hold. If $\sigma = ([1], \varnothing)$, there exists a rational column vector $b \in \mathbb{Q}^{10}$ such that

$$V_\sigma b = \big(25d(P)(d(P) - 1) - 11d(P)\big)_{P \in \mathcal{F}_{\sigma,\ell}}. \tag{61}$$

If $\sigma = ([2], \varnothing)$, there exists a rational column vector $b \in \mathbb{Q}^{54}$ such that

$$V_\sigma b = \big(25c(P)(c(P) - 1) - 14c(P) + 3d(P)\big)_{P \in \mathcal{F}_{\sigma,\ell}}. \tag{62}$$

The proof of Lemma 5 is entirely computational; see `programs/verify_equality.py`.

We also use the following uniqueness theorem.

Lemma 6 ([9, Theorem 6.4])**.** *Up to isomorphism, the Higman–Sims graph is the unique triangle-free* 22*-regular graph on* 100 *vertices in which any two distinct nonadjacent vertices have exactly* 6 *common neighbors.*

In the original notation, this graph is $r(2, 6, 22, 100)$. The parameters specify diameter 2, exactly 6 paths of length 2 between vertices at distance 2, degree 22, and order 100. The conditions in Lemma 6 imply diameter 2; they also imply girth 4, since two nonadjacent vertices and two of their common neighbors form a 4-cycle, while triangles are excluded.

Proof of the equality statement in Theorem 1. As shown in [4], for every integer $m \geq 1$, the graph HS^m is triangle-free and satisfies

$$|V(\mathrm{HS}^m)| = 100m, \qquad q_{\min}(\mathrm{HS}^m) = 14m = \frac{7}{50}|V(\mathrm{HS}^m)|.$$

Graph isomorphisms preserve the signless Laplacian spectrum, so the same equality holds for every $G \cong \mathrm{HS}^m$.

Conversely, suppose that $q_{\min}(G) = 7n/50$. Write $V = V(G)$, $A = A_G$, and

$$Q_G = \mathrm{diag}_{v \in V}(\deg_G(v)) + A.$$

Every eigenvalue of $Q_G/n - 7I/50$ is nonnegative. By (12), for every real column vector $u \in \mathbb{R}^5$,

$$u^{\mathsf{T}} \mathsf{F} u = \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U^{xy}u)^{\mathsf{T}}\left(\frac{Q_G}{n} - \frac{7}{50}I\right)(U^{xy}u)\right] \geq 0.$$

Thus F is positive semidefinite. Let B be as in the proof of (1). Hence $\mathrm{tr}(B\mathsf{F}) \geq 0$. By the proof of (1), $\mathrm{tr}(B\mathsf{F}) \leq 0$. Therefore

$$\mathrm{tr}(B\mathsf{F}) = 0. \tag{63}$$

For each type $\sigma \in \mathcal{T}_k$, $0 \leq k \leq 6$, set $\ell = \lfloor (8+k)/2 \rfloor$ and define the column vector

$$p_\sigma(\boldsymbol{r}) = \left(\Pr(P \mid \boldsymbol{r})\right)_{P \in \mathcal{F}_{\sigma,\ell}} \qquad (\boldsymbol{r} \in V^k).$$

In the following expectation, $\boldsymbol{r}$ is uniformly distributed on V^k. By (63), (57), (43), and (59), we obtain

$$\begin{aligned}
0 &= \operatorname{tr}(B\mathsf{F}) \\
&= -\sum_{k=0}^{6}\sum_{\sigma\in\mathcal{T}_k} \operatorname{tr}\left(M_\sigma \sum_{H\in\mathcal{T}_8} \Pr(G_{\boldsymbol{X}} \cong H) C_H^\sigma\right) \\
&= -8!\sum_{k=0}^{6}\sum_{\sigma\in\mathcal{T}_k} \operatorname{tr}\Big(M_\sigma \mathbb{E}_{\boldsymbol{r}}\Big[\mathbf{1}_{G_{\boldsymbol{r}}=\sigma}\big(\Pr(P \mid \boldsymbol{r})\Pr(P' \mid \boldsymbol{r})\big)_{P,P'\in\mathcal{F}_{\sigma,\ell}}\Big]\Big) \\
&= -\sum_{k=0}^{6}\sum_{\sigma\in\mathcal{T}_k} \frac{2^{80}}{a_\sigma^2} \operatorname{tr}\Big(V_\sigma(qI+X_\sigma)V_\sigma^{\mathsf{T}} \mathbb{E}_{\boldsymbol{r}}\Big[\mathbf{1}_{G_{\boldsymbol{r}}=\sigma}\big(\Pr(P \mid \boldsymbol{r})\Pr(P' \mid \boldsymbol{r})\big)_{P,P'\in\mathcal{F}_{\sigma,\ell}}\Big]\Big) \\
&= -\sum_{k=0}^{6}\sum_{\sigma\in\mathcal{T}_k} \frac{2^{80}}{a_\sigma^2} \operatorname{tr}\Big(V_\sigma(qI+X_\sigma)V_\sigma^{\mathsf{T}} \mathbb{E}_{\boldsymbol{r}}\Big[\mathbf{1}_{G_{\boldsymbol{r}}=\sigma} p_\sigma(\boldsymbol{r}) p_\sigma(\boldsymbol{r})^{\mathsf{T}}\Big]\Big) \\
&= -\sum_{k=0}^{6}\sum_{\sigma\in\mathcal{T}_k} \frac{2^{80}}{a_\sigma^2} \mathbb{E}_{\boldsymbol{r}}\Big[\mathbf{1}_{G_{\boldsymbol{r}}=\sigma} \operatorname{tr}\Big(V_\sigma(qI+X_\sigma)V_\sigma^{\mathsf{T}} p_\sigma(\boldsymbol{r}) p_\sigma(\boldsymbol{r})^{\mathsf{T}}\Big)\Big] \\
&= -\sum_{k=0}^{6}\sum_{\sigma\in\mathcal{T}_k} \frac{2^{80}}{a_\sigma^2} \mathbb{E}_{\boldsymbol{r}}\Big[\mathbf{1}_{G_{\boldsymbol{r}}=\sigma} p_\sigma(\boldsymbol{r})^{\mathsf{T}} V_\sigma(qI+X_\sigma)V_\sigma^{\mathsf{T}} p_\sigma(\boldsymbol{r})\Big]. \qquad (64)
\end{aligned}$$

Every $qI + X_\sigma$ is positive definite by condition (iii). Since all summands inside these finite averages are nonnegative and each tuple has probability $n^{-k} > 0$, it follows that for every $\boldsymbol{r} \in V^k$ with $G_{\boldsymbol{r}} = \sigma$,

$$V_\sigma^{\mathsf{T}} p_\sigma(\boldsymbol{r}) = 0. \qquad (65)$$

Consequently, for every $\boldsymbol{r} \in V^k$ with $G_{\boldsymbol{r}} = \sigma$ and every column vector w of the appropriate dimension,

$$\sum_{P\in\mathcal{F}_{\sigma,\ell}} (V_\sigma w)_P \Pr(P \mid \boldsymbol{r}) = w^{\mathsf{T}} V_\sigma^{\mathsf{T}} p_\sigma(\boldsymbol{r}) = 0. \qquad (66)$$

Fix $x \in V$ and take $\sigma = ([1], \varnothing)$. For $P \in \mathcal{F}_{\sigma,4}$, we write $\Pr(P \mid x)$ for $\Pr(P \mid (x))$ in (41), viewing (x) as a one-dimensional root vector. Let $\boldsymbol{Y} = (Y_2, Y_3, Y_4)$ be uniformly distributed on V^3. Then

$$\begin{aligned}
&\sum_{P\in\mathcal{F}_{\sigma,4}} d(P)\Pr(P \mid x) \\
&= \sum_{P\in\mathcal{F}_{\sigma,4}} d(P) \frac{\big|\{\boldsymbol{Y} \in V^3 : (G_{(x,\boldsymbol{Y})}, \iota) \cong_\sigma P\}\big|}{n^3} \\
&= \sum_{P\in\mathcal{F}_{\sigma,4}} d(P) \mathbb{E}_{\boldsymbol{Y}}\big[\mathbf{1}_{\{(G_{(x,\boldsymbol{Y})},\iota)\cong_\sigma P\}}\big] \\
&= \mathbb{E}_{\boldsymbol{Y}}\big[d(G_{(x,\boldsymbol{Y})})\big] \\
&= \mathbb{E}_{\boldsymbol{Y}}\left[\sum_{i=2}^{4} A_{xY_i}\right] \\
&= \sum_{i=2}^{4} \mathbb{E}_{Y_i} A_{xY_i} = \frac{3}{n}\sum_{z\in V} A_{xz} = \frac{3\deg_G(x)}{n}. \qquad (67)
\end{aligned}$$

Using $A_{xY_i}^2 = A_{xY_i}$, we also have

$$\begin{aligned}
&\sum_{P\in\mathcal{F}_{\sigma,4}} d(P)(d(P)-1)\Pr(P \mid x) \\
&= \sum_{P\in\mathcal{F}_{\sigma,4}} d(P)(d(P)-1)\frac{\left|\{\boldsymbol{Y}\in V^3 : (G_{(x,\boldsymbol{Y})},\iota)\cong_\sigma P\}\right|}{n^3} \\
&= \sum_{P\in\mathcal{F}_{\sigma,4}} d(P)(d(P)-1)\mathbb{E}_{\boldsymbol{Y}}\left[\mathbf{1}_{\{(G_{(x,\boldsymbol{Y})},\iota)\cong_\sigma P\}}\right] \\
&= \mathbb{E}_{\boldsymbol{Y}}\left[d(G_{(x,\boldsymbol{Y})})(d(G_{(x,\boldsymbol{Y})})-1)\right] \\
&= \mathbb{E}_{\boldsymbol{Y}}\left[\left(\sum_{i=2}^{4} A_{xY_i}\right)^2 - \sum_{i=2}^{4} A_{xY_i}\right] \\
&= \sum_{\substack{2\le i,j\le 4\\ i\neq j}} \mathbb{E}_{Y_i,Y_j}[A_{xY_i}A_{xY_j}] \\
&= \sum_{\substack{2\le i,j\le 4\\ i\neq j}} \left(\frac{1}{n}\sum_{z\in V} A_{xz}\right)\left(\frac{1}{n}\sum_{z'\in V} A_{xz'}\right) = 6\left(\frac{\deg_G(x)}{n}\right)^2.
\end{aligned} \tag{68}$$

Taking $w = b$ in (66), and using (61), (67), and (68), we obtain

$$\begin{aligned}
0 &= \sum_{P\in\mathcal{F}_{\sigma,4}} (V_\sigma b)_P \Pr(P \mid x) \\
&= \sum_{P\in\mathcal{F}_{\sigma,4}} \left(25d(P)(d(P)-1) - 11d(P)\right)\Pr(P \mid x) \\
&= 25\cdot 6\left(\frac{\deg_G(x)}{n}\right)^2 - 11\cdot\frac{3\deg_G(x)}{n} \\
&= 150\frac{\deg_G(x)}{n}\left(\frac{\deg_G(x)}{n} - \frac{11}{50}\right).
\end{aligned}$$

The equality $q_{\min}(G) = 7n/50 > 0$ excludes isolated vertices, so $\deg_G(x) > 0$. It follows that G is regular, with

$$\deg_G(x) = \frac{11n}{50} \qquad (x\in V). \tag{69}$$

We next determine the common-neighbor counts. Write $N_G(x) = \{z\in V : A_{xz} = 1\}$. For $x, y\in V$ with $A_{xy} = 0$, take $\sigma = ([2],\varnothing)$. Here $\Pr(P \mid x, y)$ denotes $\Pr(P \mid (x,y))$ in (41), with root vector (x,y). For $\boldsymbol{Y} = (Y_3, Y_4, Y_5)$ uniformly distributed on V^3,

$$\begin{aligned}
&\sum_{P\in\mathcal{F}_{\sigma,5}} c(P)\Pr(P \mid x, y) \\
&= \sum_{P\in\mathcal{F}_{\sigma,5}} c(P)\frac{\left|\{\boldsymbol{Y}\in V^3 : (G_{(x,y,\boldsymbol{Y})},\iota)\cong_\sigma P\}\right|}{n^3} \\
&= \sum_{P\in\mathcal{F}_{\sigma,5}} c(P)\mathbb{E}_{\boldsymbol{Y}}\left[\mathbf{1}_{\{(G_{(x,y,\boldsymbol{Y})},\iota)\cong_\sigma P\}}\right] \\
&= \mathbb{E}_{\boldsymbol{Y}}\left[c(G_{(x,y,\boldsymbol{Y})})\right]
\end{aligned}$$

$$
\begin{aligned}
&= \mathbb{E}_{\boldsymbol{Y}}\left[\sum_{i=3}^{5} A_{xY_i}A_{yY_i}\right] \\
&= \sum_{i=3}^{5} \mathbb{E}_{Y_i}[A_{xY_i}A_{yY_i}] = \frac{3}{n}\sum_{z\in V} A_{xz}A_{yz} \\
&= \frac{3|N_G(x)\cap N_G(y)|}{n}.
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
&\sum_{P\in\mathcal{F}_{\sigma,5}} c(P)(c(P)-1)\Pr(P \mid x,y) \\
&= \sum_{P\in\mathcal{F}_{\sigma,5}} c(P)(c(P)-1)\frac{\left|\{\boldsymbol{Y}\in V^3 : (G_{(x,y,\boldsymbol{Y})},\iota)\cong_\sigma P\}\right|}{n^3} \\
&= \sum_{P\in\mathcal{F}_{\sigma,5}} c(P)(c(P)-1)\mathbb{E}_{\boldsymbol{Y}}\left[\mathbf{1}_{\{(G_{(x,y,\boldsymbol{Y})},\iota)\cong_\sigma P\}}\right] \\
&= \mathbb{E}_{\boldsymbol{Y}}\left[c(G_{(x,y,\boldsymbol{Y})})(c(G_{(x,y,\boldsymbol{Y})})-1)\right] \\
&= \mathbb{E}_{\boldsymbol{Y}}\left[\left(\sum_{i=3}^{5} A_{xY_i}A_{yY_i}\right)^2 - \sum_{i=3}^{5} A_{xY_i}A_{yY_i}\right] \\
&= \sum_{\substack{3\le i,j\le 5\\ i\neq j}} \mathbb{E}_{Y_i,Y_j}[A_{xY_i}A_{yY_i}A_{xY_j}A_{yY_j}] \\
&= \sum_{\substack{3\le i,j\le 5\\ i\neq j}} \left(\frac{1}{n}\sum_{z\in V} A_{xz}A_{yz}\right)\left(\frac{1}{n}\sum_{z'\in V} A_{xz'}A_{yz'}\right) \\
&= 6\left(\frac{|N_G(x)\cap N_G(y)|}{n}\right)^2.
\end{aligned}
$$

For the degree count in the same flags,

$$
\begin{aligned}
&\sum_{P\in\mathcal{F}_{\sigma,5}} d(P)\Pr(P \mid x,y) \\
&= \sum_{P\in\mathcal{F}_{\sigma,5}} d(P)\frac{\left|\{\boldsymbol{Y}\in V^3 : (G_{(x,y,\boldsymbol{Y})},\iota)\cong_\sigma P\}\right|}{n^3} \\
&= \sum_{P\in\mathcal{F}_{\sigma,5}} d(P)\mathbb{E}_{\boldsymbol{Y}}\left[\mathbf{1}_{\{(G_{(x,y,\boldsymbol{Y})},\iota)\cong_\sigma P\}}\right] \\
&= \mathbb{E}_{\boldsymbol{Y}}\left[d(G_{(x,y,\boldsymbol{Y})})\right] \\
&= \mathbb{E}_{\boldsymbol{Y}}\left[\sum_{i=3}^{5} A_{xY_i}\right] \\
&= \sum_{i=3}^{5} \mathbb{E}_{Y_i}A_{xY_i} = \frac{3}{n}\sum_{z\in V} A_{xz} = \frac{3\deg_G(x)}{n} = \frac{33}{50}.
\end{aligned}
$$

Using (62) in (66), we therefore obtain

$$\begin{aligned}
0 &= \sum_{P\in\mathcal{F}_{\sigma,5}} \left(25c(P)(c(P)-1) - 14c(P) + 3d(P)\right)\Pr(P \mid x,y) \\
&= 25\cdot 6\left(\frac{|N_G(x)\cap N_G(y)|}{n}\right)^2 - 14\cdot 3\frac{|N_G(x)\cap N_G(y)|}{n} + 3\cdot\frac{33}{50} \\
&= 150\left(\frac{|N_G(x)\cap N_G(y)|}{n}\right)^2 - 42\frac{|N_G(x)\cap N_G(y)|}{n} + \frac{99}{50} \\
&= 150\left(\frac{|N_G(x)\cap N_G(y)|}{n} - \frac{3}{50}\right)\left(\frac{|N_G(x)\cap N_G(y)|}{n} - \frac{11}{50}\right).
\end{aligned}$$

Hence

$$|N_G(x)\cap N_G(y)| \in \left\{\frac{3n}{50}, \frac{11n}{50}\right\} \qquad (A_{xy}=0). \tag{70}$$

For adjacent x, y, the common-neighbor count is zero by triangle-freeness.

To recover the structure of G, partition V into the classes

$$W_x = \{y \in V : N_G(y) = N_G(x)\}.$$

Each class is independent, since adjacent vertices cannot have equal neighbors. Moreover, for $x' \in W_x$ and $y' \in W_y$,

$$A_{x'y'} = A_{xy'} = A_{xy},$$

so two distinct classes induce either a complete bipartite graph or no edges. Put $d = 11n/50$. By regularity, all neighborhoods have size d, and thus

$$|N_G(x)\cap N_G(y)| = d \quad\Longleftrightarrow\quad y \in W_x.$$

For $x \in V$,

$$\begin{aligned}
d^2 &= \sum_{z\in N_G(x)} \deg_G(z) = \sum_{z\in N_G(x)}\sum_{y\in V} A_{zy} \\
&= \sum_{y\in V}\sum_{z\in N_G(x)} A_{zy} = \sum_{y\in V} |N_G(x)\cap N_G(y)| \\
&= \sum_{y\in W_x} d + \sum_{y\in N_G(x)} 0 + \sum_{y\in V\setminus(W_x\cup N_G(x))} \frac{3n}{50} \\
&= |W_x|d + (n - |W_x| - d)\frac{3n}{50} \\
&= \frac{8n}{50}|W_x| + \frac{117n^2}{2500}.
\end{aligned} \tag{71}$$

Hence

$$|W_x| = \frac{d^2 - 117n^2/2500}{8n/50} = \frac{4n^2/2500}{8n/50} = \frac{n}{100}. \tag{72}$$

Thus $\frac{n}{100}$ is a positive integer and there are exactly 100 classes.

Let R be the graph whose vertices are these classes, with two classes adjacent if all edges between them are present in G. The classes are independent and all have size $\frac{n}{100}$, so $G \cong R^{n/100}$. The graph R is triangle-free, since any triangle in R would lift to a triangle in G, and each vertex of R has degree

$$\frac{d}{n/100} = \frac{11n/50}{n/100} = 22.$$

If W_x, W_y are distinct nonadjacent vertices of R, then x, y are nonadjacent in G and have different neighborhoods. By (70), they have exactly $3n/50$ common neighbors in G. These common neighbors form a disjoint union of classes of size $\frac{n}{100}$, so

$$|N_R(W_x) \cap N_R(W_y)| = \frac{|N_G(x) \cap N_G(y)|}{n/100} = \frac{3n/50}{n/100} = 6.$$

Lemma 6 now gives $R \cong \mathrm{HS}$. Therefore $G \cong R^{n/100} \cong \mathrm{HS}^{n/100}$, completing the proof. □

References

[1] J. Balogh, F. C. Clemen, B. Lidický, S. Norin and J. Volec, *The spectrum of triangle-free graphs*, SIAM J. Discrete Math. **37** (2023), 1173–1179. doi:10.1137/22M150767X.

[2] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, Cambridge, 2004. doi:10.1017/CBO9780511804441.

[3] S. Brandt, *The local density of triangle-free graphs*, Discrete Math. **183** (1998), 17–25. doi:10.1016/S0012-365X(97)00074-5.

[4] L. de Lima, V. Nikiforov and C. Oliveira, *The clique number and the smallest Q-eigenvalue of graphs*, Discrete Math. **339** (2016), 1744–1752. doi:10.1016/j.disc.2016.02.002.

[5] L. Lovász, *Large Networks and Graph Limits*, American Mathematical Society Colloquium Publications, vol. 60, American Mathematical Society, Providence, RI, 2012. doi:10.1090/coll/060.

[6] A. A. Razborov, *Flag algebras*, J. Symbolic Logic **72** (2007), 1239–1282. doi:10.2178/jsl/1203350785.

[7] R. S. Varga, *Geršgorin and His Circles*, Springer Series in Computational Mathematics, vol. 36, Springer, Berlin, 2004. doi:10.1007/978-3-642-17798-9.

[8] Q. Zhou, *The least signless Laplacian eigenvalue of* $\{C_3, C_5\}$*-free graphs*, preprint (2026), arXiv:2610.00875.

[9] A. Gewirtz, *Graphs with maximal even girth*, Canadian Journal of Mathematics **21** (1969), 915–934. Theorem 6.4, p. 933. https://doi.org/10.4153/CJM-1969-101-9.